

Problem 2.1 (MATLAB)

Model check for multivariate normality

The file '**dataset_problem_2_1.mat**' contains the data matrix for a random sample of 5-dimensional vector observations from a population possibly being multivariate normal.

Perform a model check for multivariate normality:

- Calculate the descriptive statistics $\bar{x}$, S , R
- Make plots of the multivariate scatter matrix plus marginal histograms and boxplots
- Calculate the squared sample Mahalanobis distances, $d_i^2 = (x_i - \bar{x})^T S^{-1} (x_i - \bar{x})$, and make a QQ-plot versus the relevant χ^2 distribution for check of approximate linearity
- Conclude

Problem 2.2 (MATLAB)

Generation of general multivariate normal distribution from standard univariate normal distribution

Illustration of the result: $Y_i \sim N_p(0, I_p) \Rightarrow X_i = \mu + \Sigma^{1/2} Y_i \sim N_p(\mu, \Sigma)$

- Define $\mu = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 9 & -3 \\ 2 & -3 & 5 \end{bmatrix}$
- Perform the eigenvalue decomposition of Σ , i.e. $\Sigma = P \Lambda P^T$
- Find $\Sigma^{1/2} = P \Lambda^{1/2} P^T$
- Generate $n = 10000$ samples of $Y_i \sim N_3(0, I_3)$ by using the univariate standard normal random generator **randn**
- Generate $X_i = \mu + \Sigma^{1/2} Y_i$, be sure to keep track of dimensions of vectors and matrices

Perform a model check for multivariate normality of the generated X_i -samples:

- Calculate the descriptive statistics $\bar{x}$, S , R
- Make plots of the multivariate scatter matrix plus marginal histograms and boxplots
- Calculate the squared sample Mahalanobis distances, $d_i^2 = (x_i - \bar{x})^T S^{-1} (x_i - \bar{x})$, and make a QQ-plot versus the relevant χ^2 distribution for check of approximate linearity
- Conclude

Problem 2.3 (Theoretical)

Properties of the multivariate normal distribution and related results

Consider a bivariate normal distribution given by $\mu = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$ $\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

- Write down the simultaneous pdf, $f(x)$, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
- Write out explicitly the (Mahalanobis) quadratic form $(x - \mu)^T \Sigma^{-1} (x - \mu)$ and simplify
- Give an expression for the 50% confidence ellipsis for the quadratic form, i.e. for the present bivariate normal distribution

Consider a random sample of multivariate vectors, $X_1, X_2, \dots, X_n$, which are independent and identically distributed as $N_p(\mu, \Sigma)$ with $n = 20$ and $p = 6$

Indicate the distributions of the following random variables, vectors and matrices

- $(X_1 - \mu)^T \Sigma^{-1} (X_1 - \mu)$
- $\bar{X}$
- $\sqrt{n}(\bar{X} - \mu)$
- $n(\bar{X} - \mu)^T \Sigma^{-1} (\bar{X} - \mu)$
- $(n - 1)S$
- $n(\bar{X} - \mu)^T S^{-1} (\bar{X} - \mu)$

Consider a random sample of multivariate vectors, $X_1, X_2, \dots, X_n$, which are independent and identically distributed as $N_p(\mu_x, \Sigma)$.

Indicate the distribution of the covariance matrix estimator

$$S_x = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^T$$

Next, consider another random sample of multivariate vectors, $Y_1, Y_2, \dots, Y_m$, which are independent and identically distributed as $N_p(\mu_y, \Sigma)$.

Indicate the distribution of the covariance matrix estimator

$$S_y = \frac{1}{m-1} \sum_{i=1}^m (Y_i - \bar{Y})(Y_i - \bar{Y})^T$$

It is desired to combine (pool) the two covariance matrix estimators for improved statistical quality (reduced estimator variance)

Indicate the distribution of the combination $(n - 1)S_x + (m - 1)S_y$

Use this to show that

$$S_p = \frac{(n-1)S_x + (m-1)S_y}{n+m-2}$$

is an unbiased, two-sample, pooled estimator of Σ

Problem 2.4 (Theoretical + MATLAB)

Illustration of Central Limit Theorem for multivariate distributions

Multivariate CLT (one version):

- $Y_1, Y_2, \dots, Y_n$ independent and identically (but not necessarily normal) distributed p -dimensional random vectors
- $E[Y_i] = \mu_Y$
- $\text{Cov}[Y_i] = \Sigma_Y$

$\Rightarrow$

- $\bar{Y} \rightarrow N_p(\mu_Y, \Sigma_Y/n)$ for $n \rightarrow \infty$
- $X \stackrel{\text{def}}{=} \sqrt{n} (\bar{Y} - \mu_Y) \rightarrow N_p(0, \Sigma_Y)$ for $n \rightarrow \infty$
- (where " $n \rightarrow \infty$ " is in practice often fulfilled for even very moderate values of n)

Illustration of multivariate CLT with example, theoretical part:

First, consider a univariate uniform distributed random variable, $Y \sim \text{Uniform}([a, b])$
Find or calculate $E[Y]$ and $\text{Var}[Y]$ (e.g. consult Wiki or use definitions of moments)

Next, consider a bivariate uniform distribution, $Y = (Y_1, Y_2) \sim 2D_Uniform([a, b], [c, d])$
with *independent* Y_1, Y_2 coordinates
Based on the above, indicate $E[Y]$ and $\Sigma = \text{Cov}[Y]$

Illustration of multivariate CLT with example, MATLAB part:

Define $[a, b] = [0, 1]$, $[c, d] = [0, 2]$, $n = 20$

Indicate $\mu_Y = E[Y]$ and $\Sigma = \text{Cov}[Y]$

- Using the notation above, generate – using ***rand*** (note: not ***randn***) - the random sample $Y_1, Y_2, \dots, Y_n$ where each $Y = (Y_1, Y_2) \sim 2D_Uniform([a, b], [c, d])$
- Calculate $\bar{Y}$
- Calculate $X \stackrel{\text{def}}{=} \sqrt{n} (\bar{Y} - \mu_Y)$

These three steps generates one bivariate observation of $X = (X_1, X_2)$.

Next do this for a big number of runs, e.g. $\text{runs} = 10000$, to generate a lot of $X = (X_1, X_2)$ observations.

Perform a model check for multivariate normality of the generated X -samples in the same way as in Problems 2.1 and 2.2 above and conclude.